

MODEM HARIKA

Hyderabad, Telangana | +91-7842362658 | harika23@iitk.ac.in

EDUCATION

Indian Institute of Technology, Kanpur

Aug 2023 – Present

Bachelor of Technology in Mechanical Engineering

CPI: 6.0/10

Relevant Coursework: Robotics, Dynamics, Control Systems, Fluid Mechanics, Thermodynamics, Mechanics of Solids, Nature and Properties of Materials, Fundamentals of Computing.

Narayana Junior College, Hyderabad (TSBIE)

2023

Class XII

97.9%

TECHNICAL SKILLS

Programming Languages: Python, C, C++

Deep Learning & CV: CNNs, ResNet50, PyTorch, TensorFlow, Keras, Data Augmentation, Preprocessing.

Optimization: AdamW, OneCycleLR (Super-Convergence), Label Smoothing, Mixed-Precision Training.

Software & Engineering: MATLAB/Simulink, AutoCAD (2D/3D), NumPy, Pandas, MS Excel, Kaggle.

Core Competencies: PID Control Tuning, SDOF Mathematical Modeling, Laplace-domain Analysis, Structural Drafting.

KEY PROJECTS

ShiftGuard10: Robust Image Classification

Mar 2026

- Developed a modified ResNet50 architecture optimized for low-resolution (32×32) images by refining initial layers to preserve spatial detail.
- Implemented Manifold Mixup and stochastic transformations to improve model robustness and prevent overfitting on small-scale data.
- Utilized OneCycleLR scheduling and AdamW optimization to achieve "Super-Convergence," ensuring faster and more stable training.
- Achieved a 0.744 Macro F1-score, demonstrating balanced classification accuracy across 10 diverse categories.

Vibration Analysis & PID Control of a Washing Machine

Feb 2026 – Mar 2026

- Developed an SDOF mathematical model and derived transfer functions via Laplace-domain analysis to evaluate system stability.
- Designed and tuned a PID controller in a Python framework, achieving a 94% reduction in vibration amplitude.
- Validated controlled vs. uncontrolled system performance against theoretical predictions for mechanical stability applications.

Applications of Large Language Models (LLMs)

Dec 2024 – Jan 2025

- Developed an OCR-based system for text translation and implemented question-answering modules for context-aware responses.
- Implemented and tuned encoder-decoder models using LSTMs and RNNs for sequence prediction tasks.
- Utilized LangChain (MapReduce and RefineChain) for efficient summarization of large-scale PDF documents and research queries.

Screw Jack Assembly Drawing (AutoCAD)

Jun 2025

- Designed a complete 3D screw jack assembly including lead screw, nut, handle, and base using engineering drawing standards.
- Generated front, top, and sectional views with precise dimensioning and layer management for professional drafting.

POSITIONS OF RESPONSIBILITY

Senior Executive (Public Relations), Udghosh, IIT Kanpur

Oct 2024

- Led guest outreach and successfully coordinated scheduling and formal invitations for high-profile personalities.

Organizer (Show Management), Udghosh, IIT Kanpur

Sep 2025 – Oct 2025

- Supervised event execution and designed creative layouts to enhance participant engagement and aesthetic quality.

SCHOLASTIC ACHIEVEMENTS

- JEE Advanced 2023:** Secured All India Rank 16,316 among 1.89 Lakh candidates.
- JEE Mains 2023:** Secured All India Rank 12,671 among 1.36 million candidates.
- TS EAMCET 2023:** Secured Telangana State Rank 823 among 1.95 Lakh candidates.
- SSC (Class X):** Achieved a perfect 10/10 GPA.